Online Appendix: Derivation of the Drift-Plus-Penalty Bound

Define the aggregate task arrival to the LEO-side queue $K_{i,k}(t)$ as

$$A^s_{i,k}(t) = \sum_{j=1}^{J} \beta_{j,k}(t) H^s_{i,j}(t). \tag{1}$$

By applying $([x]^+)^2 \le x^2$ to the task-queue and energy-queue evolution equations, the one-slot Lyapunov drift satisfies

$$\begin{aligned}
&L(\boldsymbol{\Theta}(t+1)) - L(\boldsymbol{\Theta}(t)) \le B(t) + \sum_{i=1}^{I} Q_i(t)\left[D_i(t) - U_i(t)\right] \\
&+ \sum_{i=1}^{I}\sum_{j=1}^{J} G^u_{i,j}(t)\left[C^u_{i,j}(t) - H^u_{i,j}(t)\right] \\
&+ \sum_{i=1}^{I}\sum_{j=1}^{J} G^s_{i,j}(t)\left[C^s_{i,j}(t) - H^s_{i,j}(t)\right] \\
&+ \sum_{i=1}^{I}\sum_{k=1}^{K} K_{i,k}(t)\left[A^s_{i,k}(t) - Y_{i,k}(t)\right] \\
&+ \sum_{j=1}^{J} \left(E^{\max}_j - E_j(t)\right)\left[E^{\text{cons}}_j(t) - E^{\text{solar}}_j(t)\right],
\end{aligned} \tag{2}$$

where $B(t)$ collects all quadratic terms generated by the queue and energy updates:

$$\begin{aligned}
B(t) = \frac{1}{2}\Bigg\{ &\sum_{i=1}^{I}\left[U_i^2(t) + D_i^2(t)\right] \\
&+ \sum_{i=1}^{I}\sum_{j=1}^{J}\Big[\left(H^u_{i,j}(t)\right)^2 + \left(C^u_{i,j}(t)\right)^2 \\
&\qquad + \left(H^s_{i,j}(t)\right)^2 + \left(C^s_{i,j}(t)\right)^2\Big] \\
&+ \sum_{i=1}^{I}\sum_{k=1}^{K}\Big[Y^2_{i,k}(t) + \left(A^s_{i,k}(t)\right)^2\Big] \\
&+ \sum_{j=1}^{J}\left[E^{\text{cons}}_j(t) - E^{\text{solar}}_j(t)\right]^2 \Bigg\}.
\end{aligned} \tag{3}$$

Although $B(t)$ contains instantaneous control variables such as $H^u_{i,j}(t)$, $H^s_{i,j}(t)$, $C^u_{i,j}(t)$, $C^s_{i,j}(t)$, and $E^{\text{cons}}_j(t)$, these variables are restricted by the feasible transmission, computing-resource, and energy constraints. Moreover, since $\beta_{j,k}(t)$ is binary and $H^s_{i,j}(t)$ is bounded, the aggregate arrival $A^s_{i,k}(t) = \sum_{j=1}^{J} \beta_{j,k}(t) H^s_{i,j}(t)$ is also bounded. Hence, all quadratic terms in $B(t)$ are uniformly bounded over the feasible control set. Therefore, there exists a finite constant Υ such that

$$\mathbb{E}[B(t) \mid \boldsymbol{\Theta}(t)] \le \Upsilon, \tag{4}$$

where Υ is independent of the instantaneous control variables. Taking the conditional expectation of the above drift bound and adding the penalty term $V\mathbb{E}[\sum_{i=1}^{I} T_i(t) \mid \boldsymbol{\Theta}(t)]$ yields the drift-plus-penalty upper bound used in the main paper.